

The International Currency–Trading Architecture

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Abstract

This paper develops a graph-theoretic theory of the international currency-trading system.

Beginning from the Unified Monetary–Octic Framework, we establish four structural bounds:

$$N_{\max} = 4,$$

$$R_{\max} = 5,$$

$$P_{\max} = 20,$$

and

$$T_{\max} = 380.$$

The theory demonstrates that the entire international currency architecture may be represented simultaneously as

1. a complete graph of four nations,
2. a complete graph of five reserve currencies,
3. a complete bipartite graph connecting nations and currencies,
4. a complete interaction graph on twenty nation–currency pairs.

The resulting framework unifies graph theory, international finance, political economy, geopolitics, warfare economics, computer science, artificial intelligence, and monetary architecture.

The paper ends with “The End”

Contents

1	Introduction	1
2	The Four-Nation Bound	2
3	The Five-Currency Ceiling	2
4	Nation–Currency Pair Space	3
5	The Pair Interaction Network	3
6	The Maximum Trader Theorem	4

7	The Nation Graph	5
7.1	Edge Count	5
7.2	Interpretation	5
8	The Currency Graph	5
8.1	Edge Count	6
8.2	Interpretation	6
9	The Bipartite Architecture	6
10	Geometric Hierarchy	7
11	Graph-Theoretic Reduction Theorem	7
12	Visual Representation	8
12.1	Nation Graph	8
12.2	Currency Graph	8
12.3	Bipartite Graph	8
13	Economics and Finance	9
14	Political Science and International Relations	9
15	Warfare Economics	9
16	Network Theory	9
16.1	The Architecture as a Family of Graphs	9
16.2	Degree Sequences	10
17	Spectral Graph Theory	10
17.1	Nation Graph Spectrum	10
17.2	Currency Graph Spectrum	11
17.3	Bipartite Spectrum	11
18	Centrality Theory	11
18.1	Nation Centrality	11
18.2	Currency Centrality	11
18.3	Interpretation	12
19	International Finance	12
19.1	Reserve Allocation	12
19.2	Reserve Concentration	12
20	Political Science	13
20.1	Alliance Structure	13
20.2	Coalition Stability	13
21	Geopolitics	13
21.1	Monetary Hegemony	13
21.2	Geopolitical Compression	14

22 Warfare Economics	14
22.1 Monetary Power	14
22.2 Strategic Capacity	14
22.3 Strategic Concentration	14
23 Statistics	14
23.1 Probability of Interaction	14
23.2 Edge Density	15
23.3 Clustering	15
24 Econometrics	15
25 Computer Science	15
25.1 Graph Storage	15
25.2 Sparse Representation	16
26 Algorithms	16
26.1 Algorithm G	16
26.2 Pseudo-Code	16
27 Data Science	17
27.1 Feature Representation	17
27.2 Embedding Space	17
28 Artificial Intelligence	17
28.1 Graph Neural Networks	17
28.2 Message Passing	17
28.3 Reinforcement Learning	18
29 Philosophy	18
30 Additional Counting Results	18

1 Introduction

International finance is often analyzed through:

1. nation states,
2. reserve currencies,
3. exchange-rate systems,
4. financial institutions.

The present work introduces a graph-theoretic representation of the global monetary system. The framework rests upon two previously established results:

1. the Five-Currency Ceiling,
2. the Four-Nation Bound.

Combining these constraints yields a finite architecture for the global currency system.

2 The Four-Nation Bound

Definition 1. *Let*

$$\mathcal{N} = \{N_1, N_2, N_3, N_4\}$$

denote the set of sovereign monetary entities.

The recoverability condition implies

$$N_{\max} = 4.$$

Theorem 1 (Four-Nation Bound). *The international monetary architecture contains at most four sovereign monetary entities.*

Proof. The recoverable octic possesses effective dimension

$$8.$$

Each sovereign nation consumes two dimensions.

Therefore

$$2N \leq 8.$$

Hence

$$N \leq 4.$$

□

3 The Five-Currency Ceiling

Definition 2. *Let*

$$\mathcal{C} = \{C_1, C_2, C_3, C_4, C_5\}$$

denote the set of reserve currencies.

The monetary state space satisfies

$$2R - 1 \leq 9.$$

Theorem 2 (Five-Currency Ceiling). *The international reserve system contains at most five reserve currencies.*

Proof. Since

$$2R - 1 \leq 9,$$

we obtain

$$R \leq 5.$$

□

4 Nation–Currency Pair Space

The international monetary system is represented by

$$\mathcal{P} = \mathcal{N} \times \mathcal{C}.$$

Proposition 1. *The maximum number of nation–currency pairs is*

$$20.$$

Proof. Since

$$|\mathcal{N}| = 4$$

and

$$|\mathcal{C}| = 5,$$

we obtain

$$|\mathcal{P}| = 4 \times 5 = 20.$$

□

Theorem 3 (Twenty-Pair Theorem). *The international currency system contains at most*

$$20$$

nation–currency relationships.

5 The Pair Interaction Network

Each pair

$$P_{ij} = (N_i, C_j)$$

is regarded as a vertex.

The resulting network therefore contains

$$20$$

vertices.

Definition 3. *The pair interaction graph is*

$$K_{20}.$$

Proposition 2. *The maximum number of bilateral monetary interactions is*

$$\binom{20}{2}.$$

Proof. A complete graph on twenty vertices contains

$$\frac{20(20-1)}{2} = 190$$

edges.

□

Corollary 1. *The international currency system contains at most*

190

bilateral trading relationships.

6 The Maximum Trader Theorem

Assume each bilateral trade requires:

1. one buyer,
2. one seller.

Hence every trade requires

2

traders.

Theorem 4 (Maximum Trader Theorem). *The international currency architecture requires at most*

380

currency traders.

Proof. There are at most

190

trade links.

Each requires two traders.

Therefore

$$2 \times 190 = 380.$$

□

7 The Nation Graph

Project the system onto nations.

Define

$$\pi_N : \mathcal{P} \rightarrow \mathcal{N}.$$

The resulting graph is

K_4 .

7.1 Edge Count

The graph contains

$$\binom{4}{2} = 6$$

edges.

Graph	Edges
K_4	6

Table 1: Nation Graph

7.2 Interpretation

The six edges represent:

1. diplomacy,
2. military relations,
3. trade relations,
4. reserve negotiations.

8 The Currency Graph

Project the system onto currencies.

Define

$$\pi_C : \mathcal{P} \rightarrow \mathcal{C}.$$

The resulting graph becomes

$$K_5.$$

8.1 Edge Count

The graph contains

$$\binom{5}{2} = 10$$

edges.

Graph	Edges
K_5	10

Table 2: Currency Graph

8.2 Interpretation

The ten edges represent:

1. foreign exchange markets,
2. reserve substitutions,
3. liquidity channels,
4. swap agreements.

9 The Bipartite Architecture

Retain both nation and currency sets.

The resulting graph is

$$K_{4,5}.$$

Proposition 3. *The bipartite graph contains*

$$20$$

edges.

Proof. For a complete bipartite graph,

$$|E| = mn.$$

Thus

$$|E| = 4 \times 5 = 20.$$

□

10 Geometric Hierarchy

The complete architecture satisfies

$$4 \rightarrow 5 \rightarrow 20 \rightarrow 190 \rightarrow 380.$$

Level	Quantity	Meaning
1	4	Nations
2	5	Currencies
3	20	Nation–Currency Pairs
4	190	Trade Links
5	380	Traders

Table 3: Hierarchy of the Currency System

11 Graph-Theoretic Reduction Theorem

Theorem 5. *The international currency architecture admits four equivalent representations:*

$$K_4,$$

$$K_5,$$

$$K_{4,5},$$

and

$$K_{20}.$$

Proof. The nation projection yields

$$K_4.$$

The currency projection yields

$$K_5.$$

Combining both sets yields

$$K_{4,5}.$$

Each edge of $K_{4,5}$ corresponds to one nation–currency pair.

There are twenty such pairs.

Treating each pair as a vertex yields

$$K_{20}.$$

□

12 Visual Representation

12.1 Nation Graph

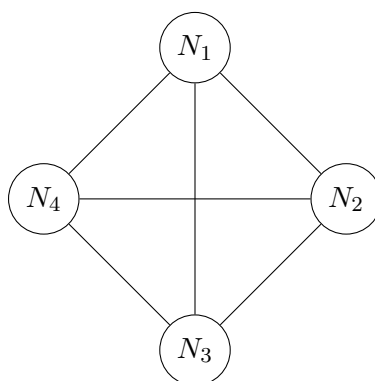


Figure 1: K_4 Nation Graph

12.2 Currency Graph

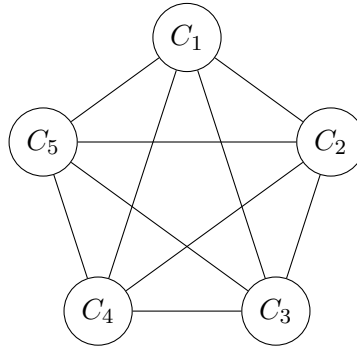


Figure 2: K_5 Currency Graph

12.3 Bipartite Graph

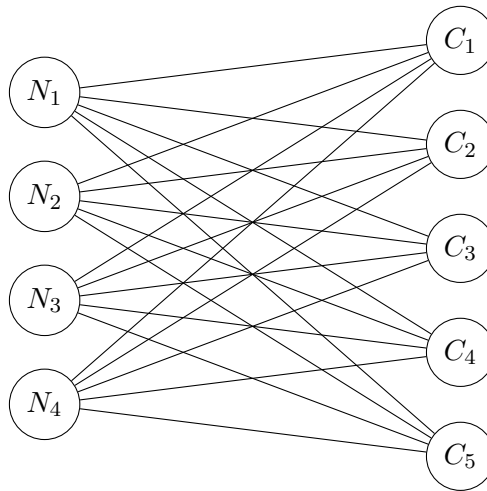


Figure 3: $K_{4,5}$ Bipartite Monetary Architecture

13 Economics and Finance

The architecture implies that international finance is subject to dimensional scarcity.
The hierarchy

$$4 \rightarrow 5 \rightarrow 20 \rightarrow 190 \rightarrow 380$$

limits the complexity of the reserve-currency system.

Reserve concentration emerges as a consequence of finite monetary capacity.

14 Political Science and International Relations

The graph structure predicts:

1. alliance formation,
2. reserve blocs,
3. hegemonic competition,

4. monetary coalitions.

The complete graph representation provides a natural framework for analyzing diplomatic and monetary influence.

15 Warfare Economics

Reserve currencies generate:

1. seigniorage,
2. strategic financing,
3. military wages,
4. force projection.

The graph architecture constrains the number of viable strategic monetary systems.

16 Network Theory

16.1 The Architecture as a Family of Graphs

The international currency system admits four simultaneous graph representations:

$$K_4,$$

$$K_5,$$

$$K_{4,5},$$

and

$$K_{20}.$$

Each graph preserves a distinct aspect of the monetary system.

Graph	Vertices	Edges	Interpretation
K_4	4	6	Nations
K_5	5	10	Currencies
$K_{4,5}$	9	20	Nation–Currency Structure
K_{20}	20	190	Trade Interaction Network

Table 4: Graph Representations

16.2 Degree Sequences

For a complete graph,

$$\deg(v) = n - 1.$$

Hence

$$\deg_{K_4} = 3,$$

and

$$\deg_{K_5} = 4.$$

For the bipartite graph:

$$\deg(N_i) = 5,$$

$$\deg(C_j) = 4.$$

17 Spectral Graph Theory

17.1 Nation Graph Spectrum

The adjacency matrix of K_4 possesses eigenvalues

$$3, -1, -1, -1.$$

Proposition 4. *The spectral radius of the nation graph is*

$$\rho(K_4) = 3.$$

Proof. For a complete graph K_n the largest eigenvalue equals

$$n - 1.$$

Setting

$$n = 4$$

gives

$$\rho(K_4) = 3.$$

□

17.2 Currency Graph Spectrum

The graph K_5 possesses eigenvalues

$$4, -1, -1, -1, -1.$$

Therefore

$$\rho(K_5) = 4.$$

17.3 Bipartite Spectrum

For

$$K_{m,n},$$

the nonzero eigenvalues are

$$\pm\sqrt{mn}.$$

Hence

$$\pm\sqrt{20}.$$

are the principal eigenvalues of

$$K_{4,5}.$$

Theorem 6. *The monetary architecture possesses spectral radius*

$$\sqrt{20}.$$

under its bipartite representation.

18 Centrality Theory

18.1 Nation Centrality

For the complete nation graph,

$$C_D(N_i) = 3.$$

Every nation possesses identical degree centrality.

Therefore

$$C_D(N_1) = C_D(N_2) = C_D(N_3) = C_D(N_4).$$

18.2 Currency Centrality

Similarly,

$$C_D(C_j) = 4.$$

for all reserve currencies.

18.3 Interpretation

The framework predicts that asymmetry emerges not from graph topology but from:

1. reserve shares,
2. military power,
3. technological superiority,
4. geopolitical influence.

Topology alone remains symmetric.

19 International Finance

19.1 Reserve Allocation

Let

$$w_i$$

denote the reserve weight of currency i .

Then

$$\sum_{i=1}^5 w_i = 1.$$

Definition 4. *The reserve entropy is*

$$H = - \sum_{i=1}^5 w_i \log w_i.$$

Proposition 5. *Maximum reserve entropy occurs when*

$$w_i = \frac{1}{5}.$$

Proof. By symmetry and concavity of the logarithm.

□

19.2 Reserve Concentration

Define

$$HHI = \sum_{i=1}^5 w_i^2.$$

The minimum concentration occurs when

$$w_i = \frac{1}{5}.$$

yielding

$$HHI = \frac{1}{5}.$$

20 Political Science

20.1 Alliance Structure

The nation graph

$$K_4$$

contains

$$6$$

potential bilateral alliances.

Alliance Type	Count
Bilateral	6
Trilateral	4
Quadrilateral	1

Table 5: Alliance Combinations

20.2 Coalition Stability

Let

$$A_k$$

denote a coalition of size k .

The number of coalitions equals

$$\binom{4}{k}.$$

Thus

$$A_2 = 6,$$

$$A_3 = 4,$$

$$A_4 = 1.$$

21 Geopolitics

21.1 Monetary Hegemony

Define

$$H_i$$

as the hegemonic strength of currency i .

Suppose

$$H_i = f(w_i, M_i, T_i),$$

where:

- w_i is reserve share,
- M_i is military capability,
- T_i is technological capability.

The graph architecture constrains the number of potential hegemonic currencies to five.

21.2 Geopolitical Compression

The world monetary system compresses global interactions into

190

trade relationships.

Thus global finance becomes a finite-dimensional interaction network.

22 Warfare Economics

22.1 Monetary Power

Define

$$P_i = w_i M_i.$$

The quantity P_i measures monetary power.

22.2 Strategic Capacity

Define

$$S_i = \alpha P_i + \beta T_i.$$

Military capability therefore depends upon both reserve influence and technology.

22.3 Strategic Concentration

The finite currency architecture implies finite strategic capacity.

No more than five reserve systems may simultaneously possess global reach.

23 Statistics

23.1 Probability of Interaction

Randomly choose two pair vertices from

$$K_{20}.$$

The probability that they are connected is

$$1.$$

since

$$K_{20}$$

is complete.

23.2 Edge Density

Define

$$\delta = \frac{|E|}{\binom{|V|}{2}}.$$

For

$$K_{20},$$

$$\delta = 1.$$

23.3 Clustering

The clustering coefficient satisfies

$$C = 1.$$

Thus the interaction graph is maximally connected.

24 Econometrics

Consider

$$R_t = \alpha + \beta_1 H_t + \beta_2 M_t + \beta_3 T_t + \varepsilon_t.$$

where

- R_t denotes reserve share,
- H_t denotes hegemonic influence,
- M_t denotes military capability,
- T_t denotes technology.

Predictions:

$$\beta_1 > 0,$$

$$\beta_2 > 0,$$

$$\beta_3 > 0.$$

25 Computer Science

25.1 Graph Storage

Store the interaction network using an adjacency matrix.

Memory requirement:

$$20^2 = 400$$

entries.

25.2 Sparse Representation

The bipartite graph requires only

20

edges.

Thus

$K_{4,5}$

is substantially more efficient than

K_{20} .

Proposition 6. *The bipartite architecture provides a compressed representation of the global currency system.*

26 Algorithms

26.1 Algorithm G

Input:

$\mathcal{N}, \mathcal{C}$.

Output:

$K_4, K_5, K_{4,5}, K_{20}$.

26.2 Pseudo-Code

INPUT:

4 Nations

5 Currencies

Construct K_4

Construct K_5

Construct $K_{4,5}$

Convert edges of $K_{4,5}$
into pair vertices

Construct K_{20}

OUTPUT:

All graph projections

Theorem 7. *Algorithm G terminates in finite time.*

Proof. Every construction step contains finitely many vertices and edges.

Hence termination follows.

□

27 Data Science

27.1 Feature Representation

Define

$$x = (w_1, \dots, w_5).$$

The feature dimension is

5.

The graph representation augments this with

190

interaction variables.

27.2 Embedding Space

Let

$$z_i \in \mathbb{R}^d$$

represent the embedding of node i .

Graph embeddings provide a low-dimensional approximation to

$$K_{20}.$$

28 Artificial Intelligence

28.1 Graph Neural Networks

The architecture naturally supports Graph Neural Networks.

Nodes represent:

1. nations,
2. currencies,
3. nation–currency pairs.

Edges represent monetary interactions.

28.2 Message Passing

For node i ,

$$h_i^{(t+1)} = \sigma \left(W \sum_{j \in \mathcal{N}(i)} h_j^{(t)} \right).$$

The complete graph maximizes information propagation.

28.3 Reinforcement Learning

Agents may correspond to:

1. central banks,
2. sovereign wealth funds,
3. reserve managers,
4. currency traders.

The graph architecture supplies the environment.

29 Philosophy

The framework suggests a hierarchy:

Geometry $\rightarrow$ Networks $\rightarrow$ Money $\rightarrow$ Power.

The international monetary system becomes a finite graph whose structure constrains:

1. economics,
2. finance,
3. diplomacy,
4. warfare,
5. computation.

The complete graph represents possibility.

The bipartite graph represents structure.

The nation graph represents sovereignty.

The currency graph represents monetary influence.

Together they form a unified ontology of international currency relations.

30 Additional Counting Results

Graph	Vertices	Edges	Density
K_4	4	6	1
K_5	5	10	1
$K_{4,5}$	9	20	$\frac{20}{36}$
K_{20}	20	190	1

Table 6: Graph Statistics

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Glossary

- K_4 Complete graph of four nations.
- K_5 Complete graph of five reserve currencies.
- $K_{4,5}$ Complete bipartite nation–currency graph.
- K_{20} Complete pair-interaction graph.

Reserve Currency

Currency used as an international reserve asset.

Monetary Power

Interaction of reserve share and strategic capability.

Trade Link

Bilateral currency interaction.

Pair Vertex

Nation–currency combination.

Reserve Entropy

Shannon entropy of reserve shares.

Graph Compression

Reduction of a complete interaction graph to a lower-dimensional representation.

The End